Calculation 2: Phase line & Stability

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Now that you can find equilibrium solutions of a differential equation, it is time to investigate what kinds of equilibrium solutions can occur. In the next video you will learn about this.

Note: in the video an example of an unstable equilibrium point is given. However, the definition of an unstable equilibrium point is not in the video. That you can find below, in the text.

Phase line & Stability of equilibrium points

4:06 / 4:06 2.0x HD

Video ▼ Subtitles ▼ Other ▼

Subtitles (captions) in other languages than provided can be viewed at YouTube. Select your language in the CC-button of YouTube.

Autonomous differential equations

In the video you have seen how you can construct a *phase line* from the direction field of a differential equation. You can only draw a phase line when the differential equation is a so-called *autonomous* differential equation.

In an *autonomous* differential equation, none of the terms depend explicitly on the independent variable.

In our differential equation,

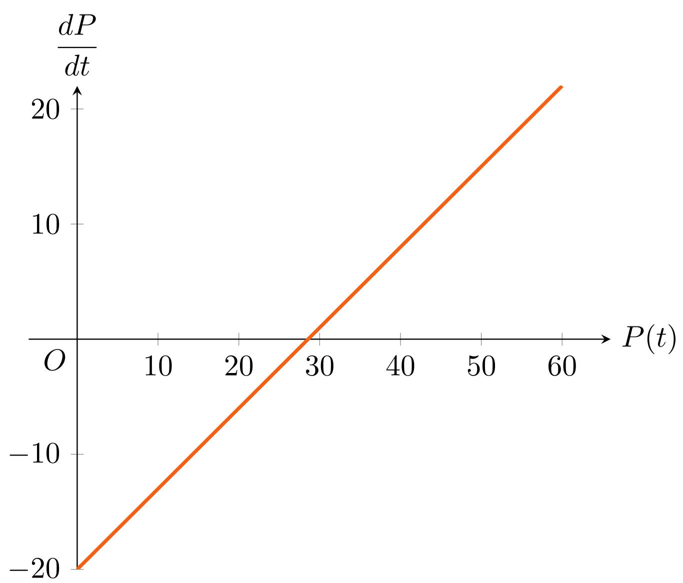
$$\frac{dP}{dt} = 0.7P(t) - 20,$$

the independent variable t does not occur explicitly: although P and $\frac{dP}{dt}$ are functions of t, the time is not mentioned in the differential equation by itself. Our equation is autonomous.

A differential equation that is not autonomous is for example $\frac{dy}{dt} = 5y(t) + \sin(t)$. For this equation you cannot draw a phase line.

Different way to draw a phase line

Because our differential equation is autonomous, you can make a graph of $\frac{dP}{dt}$ versus $P(t)$:



In this graph you can see that if $P(t)$ is smaller than the equilibrium value $\frac{20}{0.7}$, the rate change, $\frac{dP}{dt}$ is negative, so the value of $P(t)$ will decrease. This means that in the phase line you should draw an arrow pointing in the decreasing direction for values below the equilibrium. This is the same as you have seen in the video. For values of $P(t)$ larger than the equilibrium, the value of $\frac{dP}{dt}$ is positive, so the arrow in this case must be pointing in the increasing direction.

Drawing a phase line vertical takes up a lot of room, so you could also draw the phase line horizontally. For the figure above, the phase line then becomes:

Stable equilibrium point

The equilibrium solution $P_e = \frac{20}{0.7}$ is an *unstable* equilibrium solution. For this we need a definition of what a stable equilibrium is.

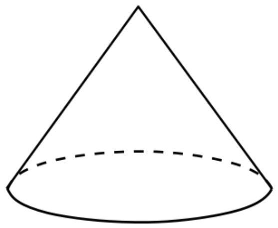
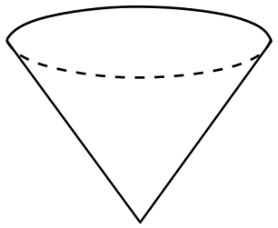
We call an equilibrium point *stable* if any initial value close to the equilibrium point gives solutions that always remain close to the equilibrium point.

For many stable equilibrium points, the solutions that start close by, do not only remain close by, but for $t \to \infty$ the solutions even tend to the equilibrium point.

Unstable equilibrium point

Any equilibrium point which is not stable we call *unstable*, so there is at least one initial value close to the equilibrium which will give a solution that moves away from the equilibrium point.

An illustration of a stable and an unstable equilibrium.

	
A cone in an stable equilibrium position. If the angle is disturbed a little, so when the cone is tilted somewhat, gravity will make the cone return to its equilibrium position.	A cone in an unstable equilibrium position. In theory, a cone can stand on its point, but any disturbance might topple it out of its equilibrium position.

Another differential equation

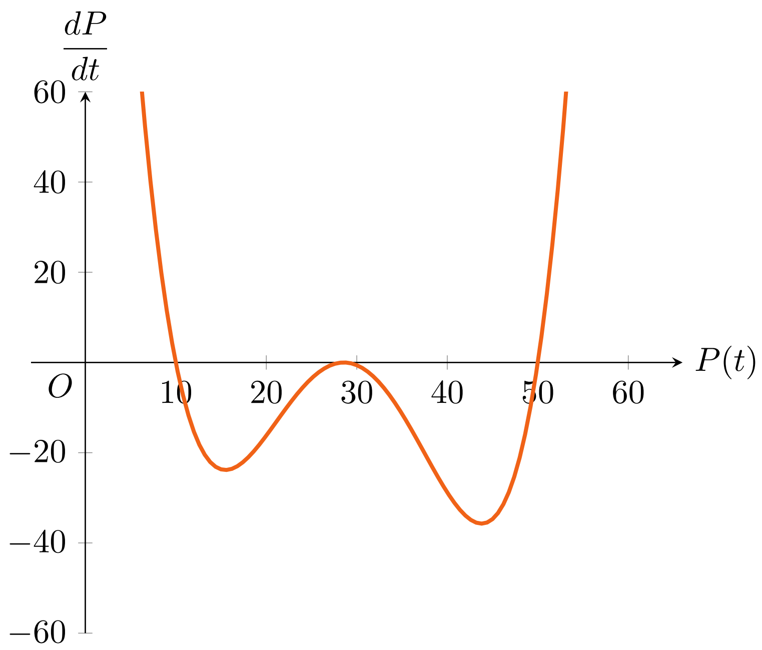
In the next exercises we will consider a "slightly" different differential equation:

$$\frac{dP}{dt} = 0.15\,(0.7P(t) - 20)^2\,(\frac{P(t)}{10} - 5)\,(\frac{P(t)}{10} - 1).$$

This differential equation has three equilibrium solutions:

$$\begin{aligned} P(t) &= 10 \\ P(t) &= \tfrac{20}{0.7} \\ P(t) &= 50 \end{aligned}$$

The graph of $\frac{dP}{dt}$ versus $P(t)$ is:

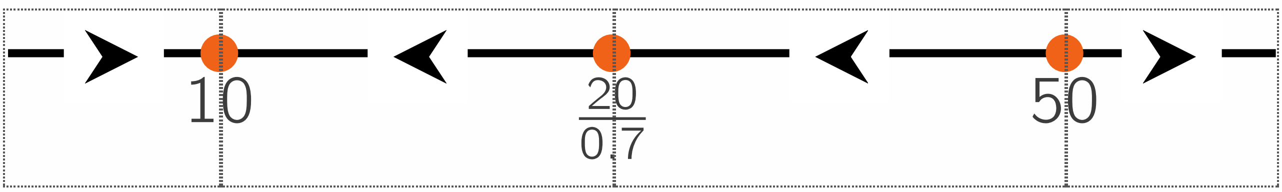


Exercise A

4/4 points (graded)

Keyboard Help

Add the correct arrow heads to the correct locations in the phase line for the differential equation above. The equilibrium points are already drawn.



Submit You have used 2 of 3 attempts.

Reset Show Answer

FEEDBACK

✔ Correctly placed 4 items.

✔ Excellent, you have correctly created the phase line.

✔ Your highest score is 4.0

Exercise B

1/1 point (graded)

The classifications stable or unstable can be derived from the arrows surrounding the equilibrium points in the phase line in exercise A.

Finish the following sentence: The equilibrium solution 10 is

- unstable, because it is surrounded by arrows pointing away from the equilibrium.
- (selected) stable, because it is surrounded by arrows pointing towards the equilibrium. ✔
- stable, because at least one of the arrows surrounding the equilibrium points towards the equilibrium.
- stable, because if $P(0) = 10$, $P(t)$ will always stay 10 and never move away.

Answer

Correct: Indeed, you can see this in the phase line you constructed in the previous exercise.

Submit You have used 1 of 2 attempts

Answers are displayed within the problem

Exercise C

1/1 point (graded)

Finish the following sentence: The equilibrium solution 50 is

- unstable, because it is surrounded by arrows pointing away from the equilibrium.
- unstable, because at least one of the arrows points away from the equilibrium. ✔
- stable, because the arrows surrounding the equilibrium both point away from the equilibrium.
- stable, because if $P(0) = 50$, $P(t)$ will always stay 50 and never move away.

Answer
Correct: Indeed, one arrow pointing away is enough to make the equilibrium unstable.

Submit You have used 1 of 2 attempts

Answers are displayed within the problem

Exercise D

1/1 point (graded)
The two classifications stable and unstable can be derived from the arrows surrounding the equilibrium points in the phase line.

Finish the following sentence: The equilibrium solution $\frac{20}{0.7}$ is

- stable, because at least one of the arrows is pointing towards the equilibrium.
- unstable, because at least one of the arrows is pointing away from the equilibrium. ✔
- neither stable, nor unstable.

Answer
Correct:
Indeed, if there is one solution that starts near the equilibrium point, and that moves away, the equilibrium point is unstable.

Submit You have used 1 of 2 attempts

Answers are displayed within the problem

In this section you have learned to sketch direction fields and phase lines. You also have seen how to find equilibrium points. Finally you have learned about the stability of the equilibrium points, which are either stable or unstable.

Now it is time to return to our original problem of predicting the number of rainbowfish in the aquarium. In the next section you will investigate how realistic the solutions of this problem are.